

Pengujian User/Profile API

Communication Protocols Demo App · HTTP/REST – JSON

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Tujuan Pengujian

Memahami komunikasi client–server pada protokol HTTP dengan pertukaran data JSON, khususnya endpoint **/api/users**.

Aplikasi demo multi-protokol:

REST HTTP :8088

MQTT :1883

gRPC HTTP/2 :50051

TLS/HTTPS :8443

★ Fokus: REST API port 8088 — HTTP/1.x plaintext dengan JSON



Skenario Bisnis

1

GET /api/users

Tampilkan seluruh data user (collection)

2

GET /api/users/1

Tampilkan detail satu user (item by id)

3

POST /api/users X

Validasi gagal (body kosong → 400)

Endpoint & Request Design

SLIDE 3

Base URL: <http://127.0.0.1:8088>

#	Method	Endpoint	Headers	Body	Expected
1	GET	/api/users	6 default	—	200 OK · JSON array + metadata
2	GET	/api/users/1	6 default	—	200 OK · JSON single object
3	POST	/api/users	7 (+Content-Type)	Body kosong	400 Bad Request · error + missing fields

Contoh body POST (field wajib: **username**, **email**, **role**): `{"username":"new.student","email":"new.student@example.edu","role":"student","status":"active"}`

Evidence GET & POST

SLIDE 4

GET Requests

GET /api/users · 200 OK

GET `http://127.0.0.1:8088/api/users`

Query Params

Key	Value	Description	Bulk Edit

Body

```
{
  "canonicalPath": "/api/users",
  "timestamp": "2025-06-09T11:08:04+0700",
  "data": [
    {
      "id": 1,
      "username": "student_demo.updated",
      "email": "student_demo.updated@example.edu",
      "role": "student",
      "status": "active",
      "department": "Sales Data",
      "updatedAt": "2025-06-09T11:08:04+0700"
    },
    {
      "id": 2,
      "username": "ops_analyst",
      "email": "ops_analyst@example.edu",
      "role": "analyst",
      "status": "active"
    }
  ]
}
```

GET /api/users/1 · 200 OK

GET `http://127.0.0.1:8088/api/users/1`

Query Params

Key	Value	Description	Bulk Edit

Body

```
{
  "id": 1,
  "username": "student_demo.updated",
  "email": "student_demo.updated@example.edu",
  "role": "student",
  "status": "active",
  "department": "Sales Data",
  "updatedAt": "2025-06-09T11:08:04+0700"
}
```

POST Request

POST /api/users · 400 Bad Request

POST `http://127.0.0.1:8088/api/users`

Query Params

Key	Value	Description	Bulk Edit

Body

```
{
  "error": "Missing required fields",
  "resource": "users",
  "missing": [
    "username",
    "email",
    "role"
  ]
}
```

Analisis Response & Encoding / Traffic

SLIDE 5

200 OK

GET /api/users
GET /api/users/1
→ Resource berhasil dikembalikan

400 Bad Request

POST /api/users (body kosong)
→ Validasi sisi server gagal
missing: username, email, role

404 Not Found

GET /favicon.ico
→ Resource tidak ada
(terlihat di Wireshark & server log)

Struktur Body Response

► Collection (/api/users)

Object + metadata + data[]

► Item (/api/users/1)

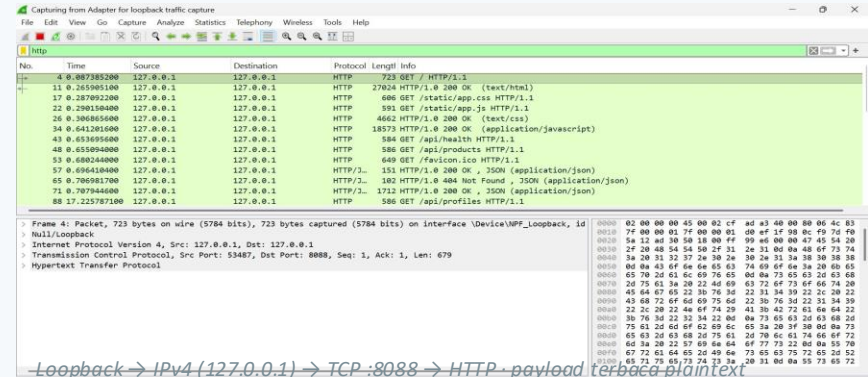
Single JSON object langsung

► Error (POST 400)

{error, resource, missing[] }

Content-Type: application/json · HTTP/1.x plaintext · UTF-8

Wireshark Capture



Kendala, Troubleshooting & Solusi

SLIDE 6

```
Command Prompt - python x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nacwa> cd C:\Users\Nacwa\Downloads\communication-protocols-demo-app\communication-protocols-demo-app

C:\Users\Nacwa\Downloads\communication-protocols-demo-app> cd C:\Users\Nacwa\Downloads\communication-protocols-demo-app\communication-protocols-demo-app
The filename, directory name, or volume label syntax is incorrect.

C:\Users\Nacwa\Downloads\communication-protocols-demo-app> python server.py
Real MQTT broker running at 127.0.0.1:1883
Wireshark capture: tcp.port == 1883; display filter: mqtt
Real gRPC HTTP/2 server running at 127.0.0.1:50051
Wireshark capture: tcp.port == 50051; Decode As HTTP2 if needed.
TLS/HTTPS demo server running at https://127.0.0.1:8443
Wireshark capture: tcp.port == 8443; display filter: tls
Communication Protocols Demo App running at http://127.0.0.1:8080
Press Ctrl+C to stop.
127.0.0.1 - "GET / HTTP/1.1" 200 -
127.0.0.1 - "GET /static/app.css HTTP/1.1" 200 -
127.0.0.1 - "GET /static/app.js HTTP/1.1" 200 -
127.0.0.1 - "GET /api/health HTTP/1.1" 200 -
127.0.0.1 - "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - "GET /api/products HTTP/1.1" 200 -
127.0.0.1 - "GET /api/profiles HTTP/1.1" 200 -
127.0.0.1 - "GET /api/profiles HTTP/1.1" 200 -
127.0.0.1 - "GET /api/profiles HTTP/1.1" 200 -
127.0.0.1 - "GET /api/users HTTP/1.1" 200 -
127.0.0.1 - "GET /api/users HTTP/1.1" 200 -
127.0.0.1 - "GET /api/users HTTP/1.1" 200 -
127.0.0.1 - "GET /api/users HTTP/1.1" 200 -
127.0.0.1 - "GET /api/users/1 HTTP/1.1" 200 -
127.0.0.1 - "POST /api/users HTTP/1.1" 400 -
```

Server log (CMD) – real-time access log

1

Error `cd` di Command Prompt

[!] Tanda kutip ganda menempel pada path → "filename syntax is incorrect"

[v] Jalankan ulang cd tanpa kutip: `cd C:\...\communication-protocols-demo-app`

2

Evidence POST 201 belum tercapture

[!] Pengujian POST hanya mendokumentasikan skenario gagal (400 Bad Request)

[v] Kirim POST dengan body lengkap {username, email, role} → tambahkan screenshot 201

3

GET /favicon.ico → 404

[!] Browser/client secara otomatis request favicon; route tidak terdaftar

[v] Bukan error kritis – server sudah mengembalikan 404 dengan JSON; cukup didokumentasikan

Temuan Utama

01

REST Konsisten

Collection (/api/users) -> array + metadata
Item (/api/users/1) -> single object

02

Performa Baik

Response time 8-11 ms (loopback)
Status 200 OK pada semua GET request

03

Validasi Server

POST tanpa field wajib -> 400 Bad Request
Error JSON terstruktur: missing[]

04

Encoding JSON

HTTP/1.x plaintext + Content-Type: application/json
Terbaca langsung di Wireshark (no TLS)